

BAO–HHT Integration: 2-bin $\rho(z)$ Update

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Overview

You tested a two-bin BAO covariance $\rho(z)$ with a split at $z=0.7$, $\rho_{lo}=+0.374$ ($z<0.7$) and $\rho_{hi}=-0.117$ ($z\geq0.7$).

Relative to the constant- ρ baseline (+0.174), the 2-bin model reduces the BAO χ^2 by ≈ 10.4 ($237.17 \rightarrow 226.77$) and the total χ^2 by ≈ 13.17 , while keeping (H_0, Ω_m) nearly unchanged.

Model comparison (constant- ρ vs 2-bin $\rho(z)$)

Model	H_0	Ω_m	χ^2_{BAO}	χ^2_{total}	AIC (eff.)	BIC (eff.)	$\Delta\chi^2_{total}$
Constant ρ	73.13554 0	0.23653 8	237.17 2	2334.68 1	2338.68 1	2349.57 6	—
2-bin $\rho(z)$ ($z<0.7$:+0.374, $z\geq0.7$:-0.117)	73.15597 4	0.23685 2	226.77 4	2321.51 4	2327.51 4	2343.85 6	-13.167

Note: For a fair comparison, AIC/BIC for the 2-bin case include a +1 hyper-parameter penalty for $\{\rho_{lo}, \rho_{hi}\}$ compared to constant- ρ .

PLamb’s printed AIC/BIC count only (H_0, Ω_m).

Per-z χ^2 contributions (2-bin – constant- ρ)

z	$\Delta\chi^2$ (2-bin – const)	Interpretation
0.380	-9.805	Improves fit at this z-pair (notably $z \approx 0.38, 0.70$).
0.510	+3.858	Slightly worse fit at this z-pair.
0.700	-9.735	Improves fit at this z-pair (notably $z \approx 0.38, 0.70$).
0.845	+2.520	Slightly worse fit at this z-pair.
0.850	-0.387	Improves fit at this z-pair (notably $z \approx 0.38, 0.70$).
1.480	-0.707	Improves fit at this z-pair (notably $z \approx 0.38, 0.70$).
2.330	+1.017	Slightly worse fit at this z-pair.

HHT interpretation & projection notes

Mapping residuals to $\log(1+z)$ for HHT: IMF “frequency” is oscillations per unit $\log(1+z)$; bandwidth reflects temporal coherence across cosmic time.

IMF energy (amplitude²) vs $\log(1+z)$ localizes which redshift ranges drive χ^2 .

In these fits, the two-bin covariance reduces χ^2 mainly near $z \approx 0.38$ and $z \approx 0.70$, consistent with LOOCV diagnostics.

This points to a mild redshift dependence in the DM–DH correlation.

Key equations / definitions (flat Λ CDM)

$$E(z) = H(z)/H_0 = \sqrt{[\Omega_m(1+z)^3 + (1-\Omega_m)]}$$

$$D_M(z) = (c/H_0) \int_0^z dz'/E(z')$$

$$D_H(z) = c/H(z) = c/(H_0 E(z))$$

Predictions: D_M/r_d and D_H/r_d ; χ^2 uses 2×2 DM–DH blocks per-z.

Two-bin covariance: $\rho(z) = \rho_{\text{lo}}$ for $z < z_{\text{split}}$, ρ_{hi} for $z \geq z_{\text{split}}$ (off-diagonal term).

Artifacts & paths

- Constant- ρ HHT export: plamb_runs/tools/bao_hht_const_rho0174/
- 2-bin HHT export (long-chain best): plamb_runs/tools/bao_hht_2bin_z0p7_longbest/
- Per- z $\Delta\chi^2$ CSV: plamb_runs/tools/bao_chi2_perz_const_vs_2bin.csv
- LOOCV (2-bin): plamb_runs/tools/bao_loocv_2bin_z0p7/bao_loocv.csv
- ρ -surface scan ($z=0.7$): plamb_runs/tools/bao_2bin_rho_surface_z0p7.csv

Recommended next checks

- Verify stability when adding/altering low- z BAO points.
- Consider a limited 3-bin test if justified by data volume.
- Cross-validate HHT energy localization vs. the $\Delta\chi^2$ map.
- Re-evaluate IC penalties if promoting $\rho(z)$ to a parametric form.